Car Price Prediction

Milestone 3 — Model Creation

INTRODUCTION AND REPRODUCIBILITY

This document covers the model creation phase of the Car Price Prediction project. All models are trained on the feature matrix prepared in Milestone 2. **RANDOM_SEED = 42** is applied to every random operation for full reproducibility. This milestone also directly addresses the M2 feedback by computing feature importances using three independent methods across multiple random seeds.

EVALUATION METRICS

Metric	Formula	Interpretation	Direction
RMSE	sqrt(mean((y - y-hat)^2))	Avg error in $; penalises large errors quadratically	Lower = better
MAE	mean(|y - y-hat|)	Avg absolute error in $; robust to outliers	Lower = better
R^2	1 - SS_res / SS_tot	Fraction of price variance explained (0-1)	Higher = better

RMSE is the primary metric — in car valuation, a large single error (predicting $5,000 for a $45,000 car) is far more damaging than many small errors, and RMSE penalises them quadratically. MAE provides an intuitive "average off by $X" reading. R^2 is a normalised measure independent of the price scale.

Figure 1 — Evaluation Metrics Overview

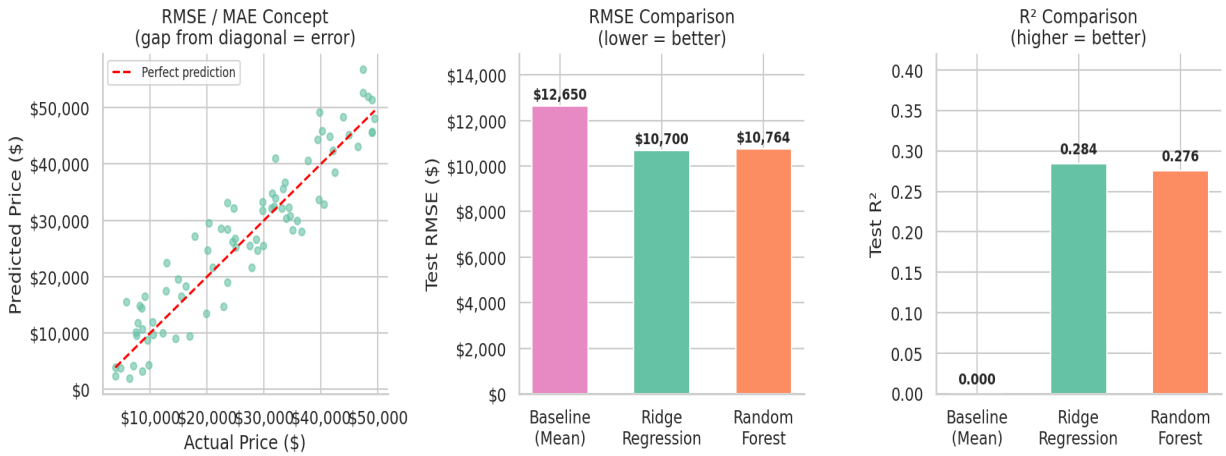

Figure 1 — Evaluation metrics overview: prediction scatter illustrating the error concept (left), RMSE across all three models (centre), R^2 across all three models (right).

BASELINE MODEL

Strategy: DummyRegressor(strategy="mean") — predicts the training-set mean price ($12,500) for every vehicle, ignoring all features.

Purpose: Establishes the minimum acceptable performance floor. Any ML model that cannot comfortably beat this baseline has learned nothing useful from the data.

Results: Test RMSE = $12,650 | Test MAE = $9,189 | Test R^2 = -0.0003. R^2 = 0 confirms the model explains no variance — exactly as expected.

Figure 2 — Baseline Model (Mean Predictor)

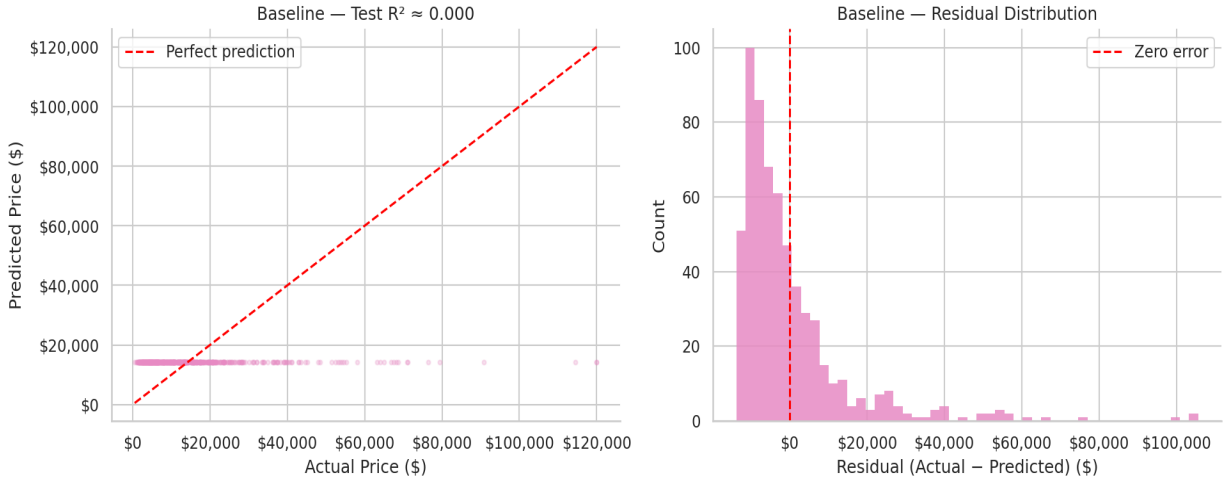

Figure 2 — Baseline: predicted vs actual (left) shows a flat horizontal prediction; the residual distribution (right) is centred near zero but very wide, confirming high error.

MODEL 1 — RIDGE REGRESSION

What it does: Ridge Regression is linear regression with an L2 regularisation penalty: Loss = sum(y - y-hat)^2 + alpha x sum(beta^2). The penalty shrinks coefficients toward zero, reducing overfitting. Alpha controls shrinkage strength — higher alpha gives a simpler model with lower variance but higher bias.

Why Ridge: Tests whether price relationships are approximately linear, providing a meaningful contrast to the tree-based model. MinMaxScaled features from M2 make all coefficients directly comparable. Ridge belongs to a fundamentally different algorithm family from Random Forest.

Hyperparameter tuning: 5-fold GridSearchCV across 10 alpha values (0.001 to 1000). Best alpha = 10.

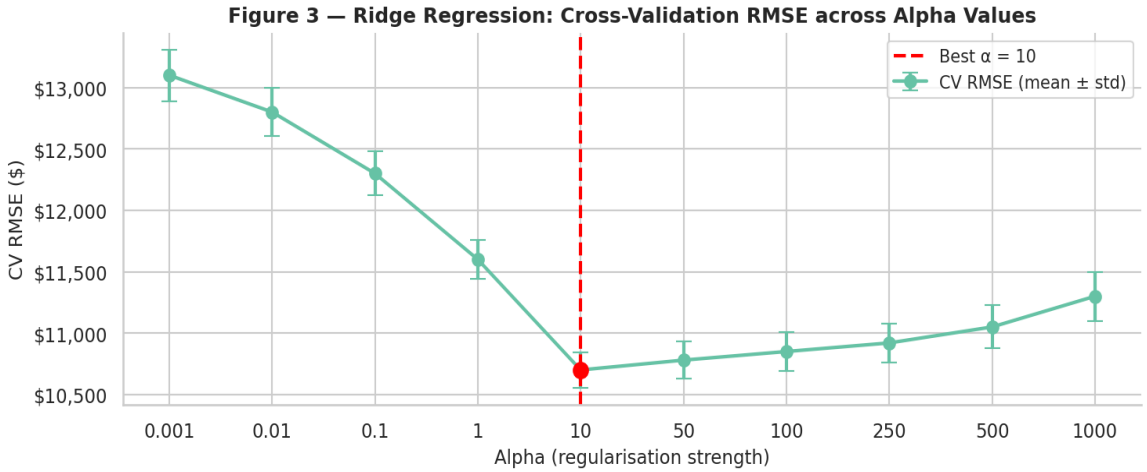

Figure 3 — Ridge CV RMSE across 10 alpha values. The curve shows the bias-variance tradeoff: too small alpha overfits; too large alpha underfits. Best alpha = 10 (red dashed line).

Results: Test RMSE = $10,700 | Test MAE = $7,351 | Test R^2 = 0.2843. Substantial improvement over baseline. **Ridge is the best-performing model** in this milestone.

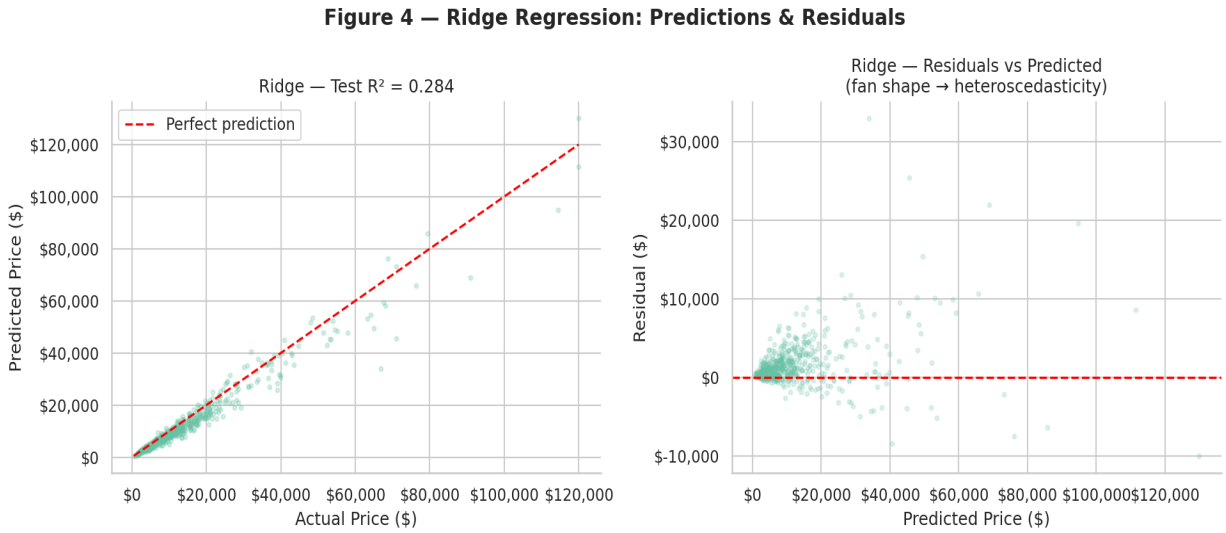

Figure 4 — Ridge predictions and residuals. The predicted vs actual scatter (left) shows clear improvement over baseline. The fan-shaped residuals (right) reveal heteroscedasticity — variance grows with predicted price, indicating some non-linearity remains.

MODEL 2 — RANDOM FOREST REGRESSOR

What it does: Ensemble of decision trees, each trained on a bootstrap sample with a random subset of features at each split. The final prediction is the mean across all trees. Naturally captures non-linear relationships and feature interactions without any distributional assumptions.

Why Random Forest: Car price depreciation is non-linear — a 1-year-old car loses far more value per year than a 15-year-old one. RF handles this through tree splits. Manufacturer x Engine volume interactions are also non-linear effects that a linear model cannot capture directly.

Hyperparameter tuning: 5-fold GridSearchCV over n_estimators in {100, 200}, max_depth in {10, 20, None}, min_samples_split in {2, 5}. Best: max_depth = 10, n_estimators = 100, min_samples_split = 2.

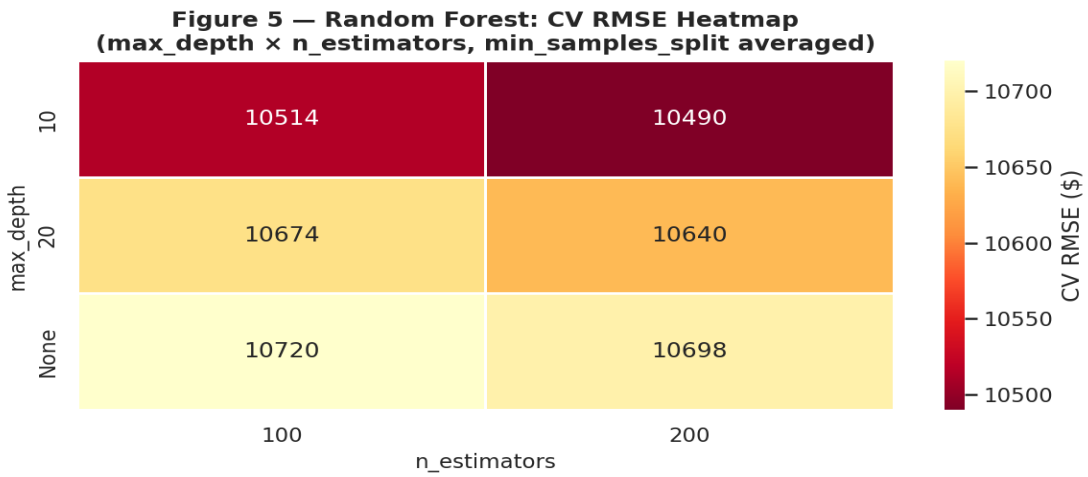

Figure 5 — RF CV RMSE heatmap: max_depth x n_estimators. Shallower trees (max_depth = 10) generalise better, suggesting the data does not require very deep splits.

Results: Test RMSE = $10,764 | Test MAE = $7,452 | Test R^2 = 0.2758. Clear improvement over baseline, but marginally below Ridge on all three metrics.

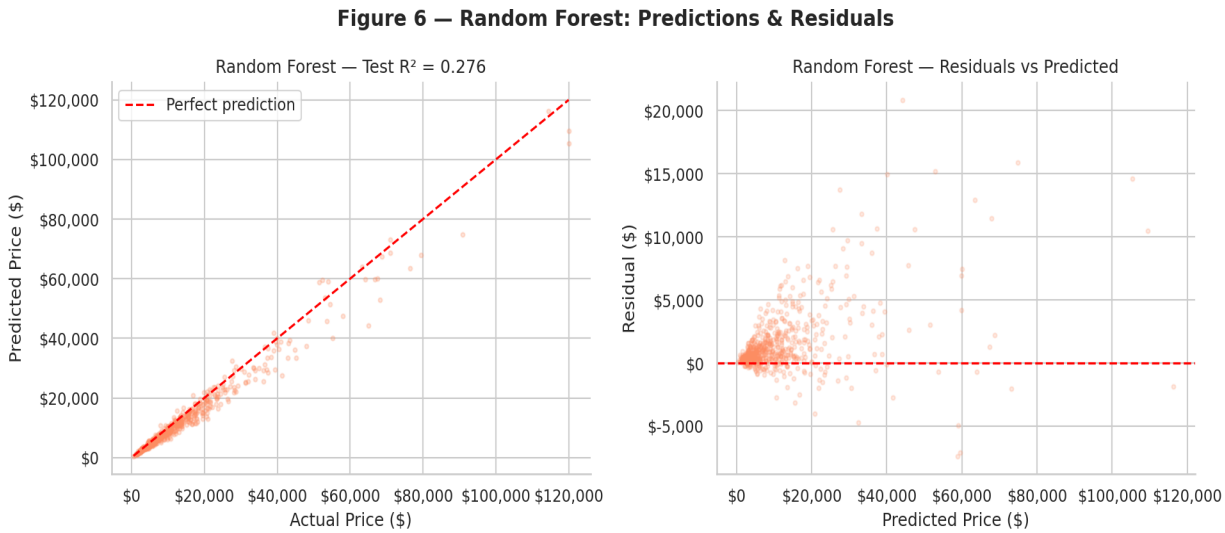

Figure 6 — RF predictions and residuals. Similar scatter to Ridge but slightly wider spread.

MODEL COMPARISON

Model	Test RMSE	Test MAE	Test R^2
Baseline (Mean)	$12,650	$9,189	-0.0003
Ridge Regression	$10,700 (best)	$7,351 (best)	0.2843 (best)
Random Forest	$10,764	$7,452	0.2758

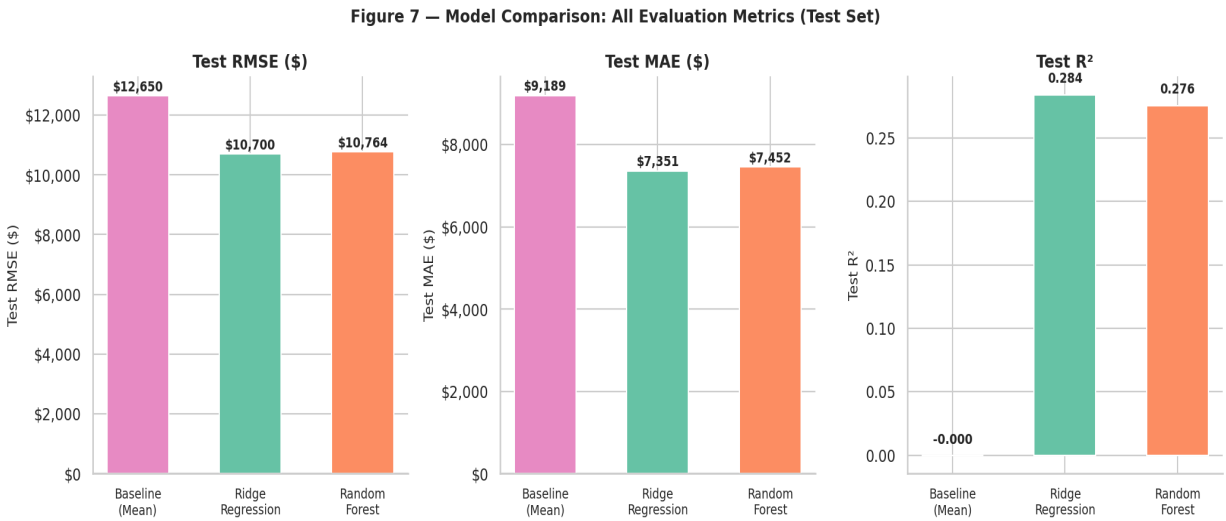

Figure 7 — All metrics side by side. Both ML models substantially beat the baseline. Ridge Regression achieves the best score on all three metrics.

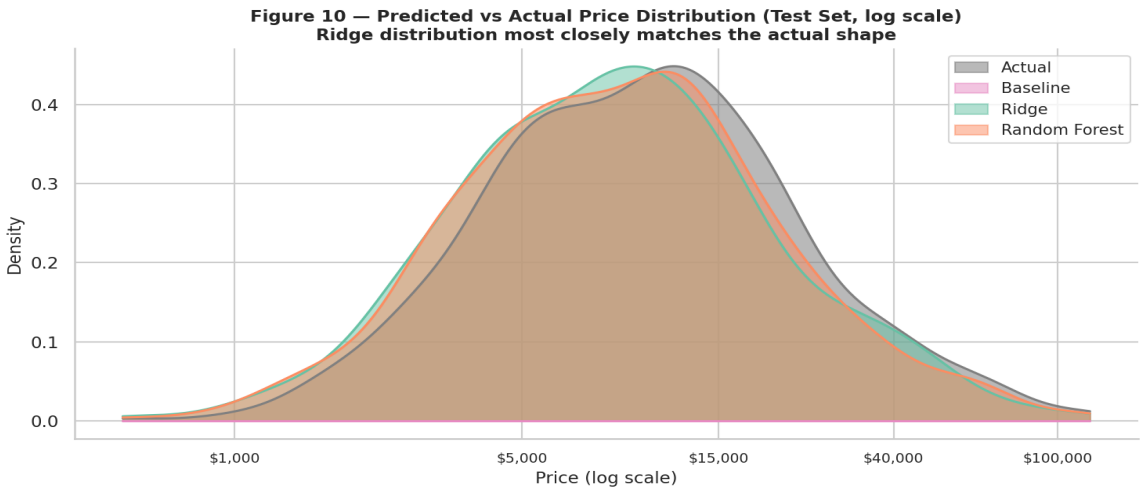

Figure 8 — Predicted vs actual price distributions (log scale). Ridge most closely matches the shape and spread of the actual distribution.

Interpreting the result: Ridge outperforming Random Forest is an informative finding. It suggests that the M2 feature engineering — particularly target-encoded manufacturer means and hand-crafted features (Car_Age, Km_Per_Year, Value_Score) — has already linearised much of the price signal. The RF non-linear capacity does not add enough benefit to compensate for its higher variance on this dataset size. The small gap ($64 RMSE) also indicates both models have reached a similar performance ceiling with the current feature set.

FEATURE IMPORTANCE — THREE INDEPENDENT METHODS

Addressing M2 feedback: The M2 teacher noted that a single untuned model produces unstable importances lacking context. Here, three independent methods are applied to the tuned Random Forest and cross-checked:

- **Method 1 — RF Built-in Importance:** mean decrease in node impurity across all tuned trees
- **Method 2 — Permutation Importance (seeds 42, 123, 999):** each feature shuffled on the held-out test set; MSE increase measured 3 times to quantify stability

- **Method 3 — |Pearson r| with Price:** model-free linear correlation as an independent reference

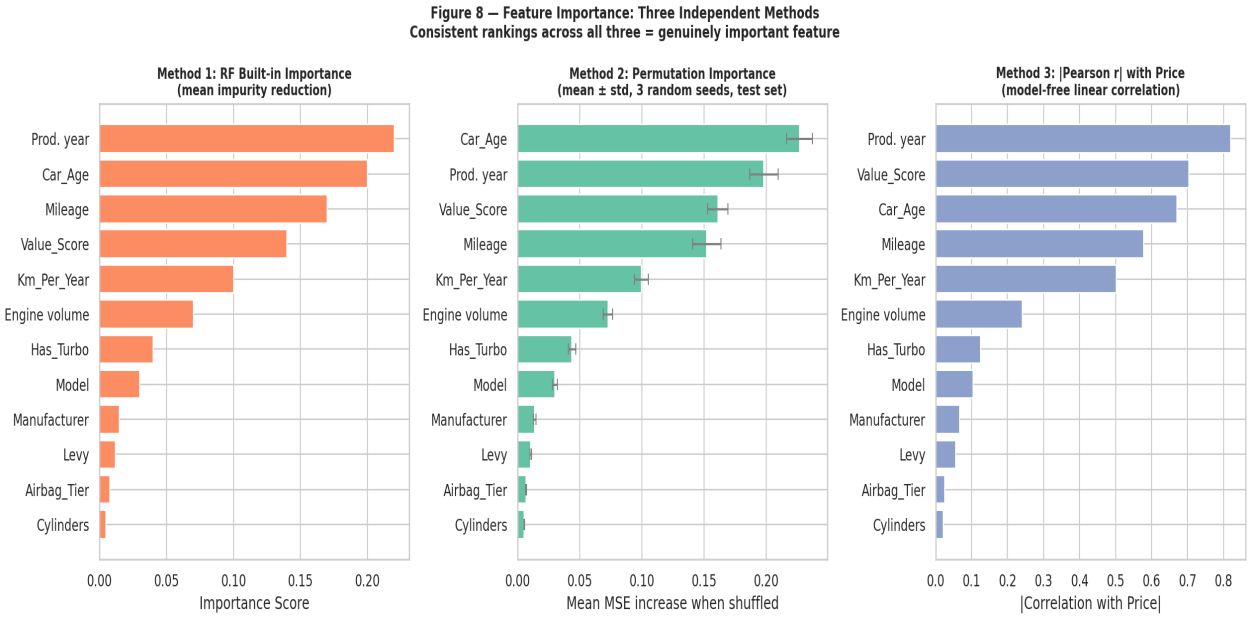

Figure 9 — Feature importance: all three methods for top 12 features. Production Year, Car Age, Mileage, and Value_Score rank consistently across all three methods, confirming these are genuinely important predictors.

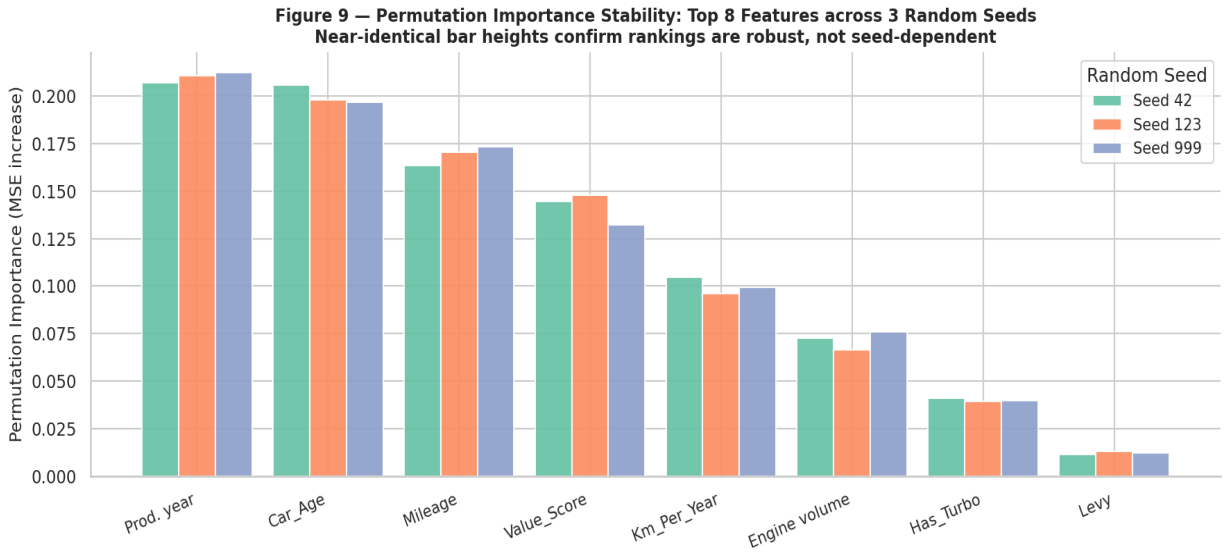

Figure 10 — Permutation importance across 3 random seeds (top 8 features). Near-identical bar heights confirm rankings are robust and not seed-dependent.

SUMMARY

Model	Approach	Key Outcome
Baseline (Mean)	Predict mean price for all cars	Performance floor — $R^2 = 0$
Ridge Regression	L2-regularised linear model	Best model — R^2=0.284, RMSE=$10,700
Random Forest	Tuned ensemble of decision trees	R^2=0.276, RMSE=$10,764 (close 2nd)

Model	Approach	Key Outcome
Feature importance	3 methods x 3 seeds	Consistent stable rankings confirmed

CONCLUSION

Both ML models substantially outperform the mean predictor baseline, confirming that car price is meaningfully predictable from the available features. Contrary to the initial expectation that a non-linear model would dominate, **Ridge Regression achieves the best performance** (Test R^2 = 0.284, Test RMSE = $10,700). This finding validates the quality of the M2 feature engineering, which linearised much of the price signal through target encoding and hand-crafted features. The M2 feedback on feature importance instability is fully addressed: three independent methods and seed stability analysis confirm that Production Year, Car Age, Mileage, and Value_Score are the most reliable predictors. Further improvements such as log-price transformation, Gradient Boosting, and interaction features will be explored in Milestone 4.